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Paleomagnetism of Anorthosite in Southern Norway and Comparison with an Equivalent in Quebec

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ABSTRACT

Paleomagnetic measurements (including alternating field demagnetization) on 16 oriented samples from five sites in the Precambrian anorthosite massif at Egersund, southern Norway, revealed a stable, consistent reversed remanent magnetization (after 150 oersted AF demagnetization, $D = 315°$, $I = -82°$, $k = 92$, $\alpha_{95} = 8°$). The calculated paleomagnetic (south) pole lies at 42° N., 20° E. and is more than 45° from the pole (39° N., 40° W.) previously obtained from anorthosite of similar age at Allard Lake, Quebec. If Eurasia is fitted to North America according to the model of Bullard and others (1965), the rotated Egersund pole is at 42° N., 18° W. Allowing for the uncertainties in the data, this is not significantly different at the 5-percent level from the Allard Lake pole, and suggests that about 1,000 m.y. ago North America and Europe were contiguous as in Laurasian reconstructions.

INTRODUCTION

Anorthosite massifs (as distinct from anorthosites in stratiform complexes) are an uncommon plutonic rock series (see compilation of known occurrences by Anderson, 1969) of generally similar petrology throughout the world. Herz (1969) gave particular significance to this similarity by showing that all the anorthosite bodies of the world have ages between 1,000 and 1,700 m.y. and that they tended to fall in two distinct belts when Laurasia and Gondwana were reassembled.

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Hargraves and Burt (1967) had obtained a paleomagnetic pole for the anorthosite series rocks at Allard Lake, Quebec. In view of the petrological and age similarity of the Quebec anorthosites to those in southern Norway, Hargraves thought to test the possibility of their predrift contiguity by obtaining paleomagnetic data from the Norwegian rocks for comparison. A suite of oriented samples was collected from the Egersund anorthosite massif in July 1968. This paper reports the paleomagnetic results obtained, and their comparison with the data from Allard Lake, Quebec.

SAMPLING AND MEASUREMENT

Sixteen hand samples were collected from fresh roadcuts at five sites in the Egersund area (Fig. 1). The samples were oriented by means of an astrocompass (La Rochelle, 1961). The remanent magnetism of at least two 1 in. long $\times$ 1 in. diameter specimen cores drilled from each sample was measured on a Princeton Applied Research SM1 spinner magnetometer. Reconnaissance AF demagnetization of selected cores up to fields of 500 oersted revealed that the anorthosites possessed NRMs varying from 10^{-2} to 5×10^{-5} emu/cc, which was, however, in all cases relatively stable (see Fig. 2). In light of these data, all the remaining cores were demagnetized at 50, 150, and 300 oersted. The mean vectors, at sample and site and locality levels, were negligibly different at 150 and 300 oersted (*see* Table 1), but the statistics for the results obtained at 150 oersted are slightly better, and the 150 oersted mean vector has been used for the pole determination.

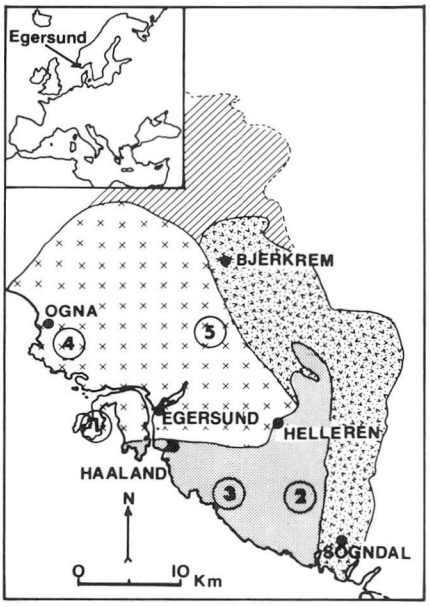


Figure 1. Map showing location of Egersund in Norway, and generalized geology of the area (modified *from* P. Michot, 1960). The two principal anorthosite bodies—Egersund-Ogna (x-pattern) and Haaland-Helleren (stippled)—are shown, and the paleomagnetic sample sites (circled numbers) within them; the v-dot pattern marks the Bjerkrem Sogndal mangerite massif; the diagonal lines signify charnockites.

PETROLOGY AND MINERALOGY AND RELATION TO MAGNETIC PROPERTIES

The geology of the Egersund area (Fig. 1) has been described in several papers by P. Michot and J. Michot. Apart from other intrusive units, they distinguish two separate anorthosite massifs, the Egersund-Ogna (P. Michot, 1960, 1969; sample sites 1, 4, and 5), and the Haaland-Helleren (J. Michot, 1961; sample sites 2 and 3). In general the massifs consist of coarse, massive, leuco-anorthosite cores, with more gneissic, noritic border zones. The anorthositic rocks are cut by somewhat younger

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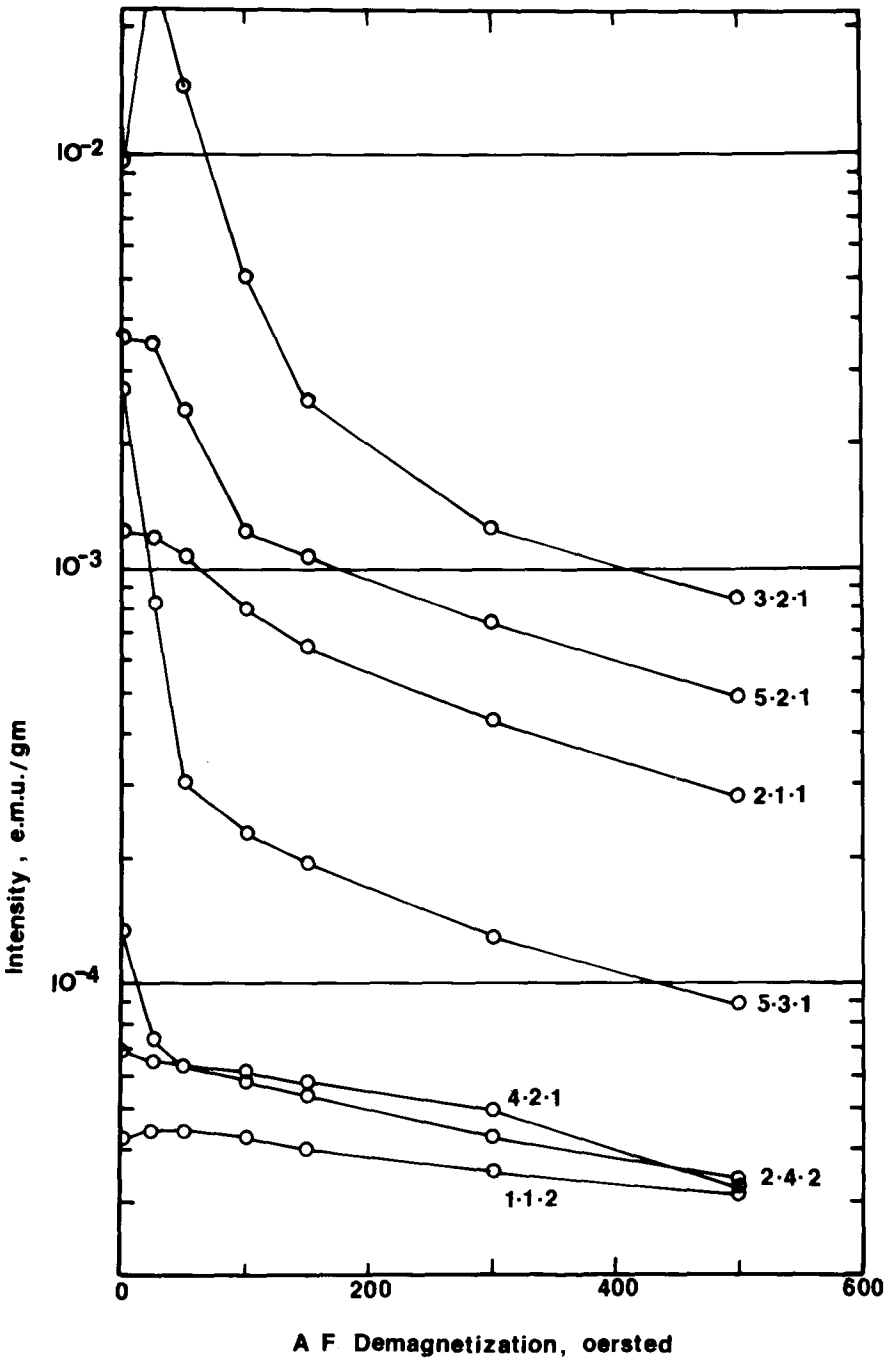


Figure 2. Alternating field demagnetization curves for representative specimens of the Egersund rocks sampled. Specimen 3-2-1 is from a pyroxene megacryst; 2-4-2 from a plagioclase megacryst; 5-3-1 is monzonorite; the others are anorthosite from sites 1, 2, 4, and 5. Note high intensity of the pyroxene and the relative instability of the monzonorite.

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TABLE 1. PALEOMAGNETIC RESULTS OBTAINED IN EGERSUND STUDY

Site	N	n	Peak Field oersted	D	I	K	$\alpha 95$
1	3	7	NRM	311.4	−75.3	44.7	18.7
			150	321.1	−78.8	173.7	9.4
			300	325.2	−77.1	192.9	8.9
2	4	9	NRM	303.1	−50.9	4.9	46.3
			150	295.1	−70.0	231.4	6.1
			300	298.6	−69.8	98.9	9.3
3	2	4	NRM	270.3	−34.7	4.1	90.0
			150	337.9	−83.2	12.7	77.2
			300	320.8	−85.9	13.0	76.1
4	3	6	NRM	58.2	−85.1	368.2	6.4
			150	64.7	−86.3	975.9	3.9
			300	71.7	−85.4	1074.9	3.8
5	4	11	NRM	0.2	−73.2	3.3	60.3
			150	311.1	−84.3	65.9	11.4
			300	298.3	−84.3	51.6	12.9
Mean of the 5 sites:			NRM	300.1	−68.3	9.8	25.8
			150	315.4	−81.9	91.9	8.0
			300	314.4	−82.1	79.1	8.7

Sites: *see* Figure 1; N = number of samples;
n = number of specimens actually measured; Peak field oersted signifies intensity of AF demagnetization;
D = declination, I = inclination; k = estimate of Fisher's dispersion parameter K; $\alpha 95$ = half angle of cone of 95% confidence.

(although judged to be consanguineous) dikes and sheets of quartz norites or monzonorites (P. Michot, 1960, p. 28). The petrological similarity between this and the North American anorthosite massifs in the Adirondacks (New York State) and Quebec has been pointed out by several authors (for example, Hargraves, 1962, p. 182).

The oriented samples used in this study are, with three exceptions, coarse-grained (up to 1 cm) leucocratic anorthosite. The rocks showed little evidence of crushing or peripheral granulation, and the plagioclase in several samples displayed the chatoyance of laboradorite. Of the three exceptional samples, one is a plagioclase megacryst, one is a hypersthene megacryst, and the third is a sample of monzonorite from a dikelike body at site 5.

In thin section, the anorthosites consist of a coarse aggregate of anhedral plagioclase with subsidiary, strongly pleochroic hypersthenes, and minor opaque oxide. The plagioclase is charged with fine rod-shaped inclusions ($2\mu \times 10\mu$) which seem to be mostly ferromagnesian silicates, but some are opaque. In some crystals the inclusions appear to have aggregated to form larger crystals, the immediate surroundings being clear.

The hypersthene has relatively poor cleavage, which in some crystals is lined with fine reddish lamellae (exsolution?), imparting a schiller appearance. In all samples examined the hypersthene crystals are traversed by prominent serpentinized cracks, with associated fine opaque minerals.

In reflected light, the accessory opaque mineral is seen to be a coarse hematite-ilmenite intergrowth. The opaque rodlike inclusions in the feldspars, when large

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enough to identify, are also intergrowths of hematite and ilmenite in approximately equal proportions. These are indentical to the inclusions in the Brulé Lake (Quebec) anorthosite-plagioclase described by Anderson (1966).

The pinkish-brown schiller lamellae in the hypersthene likewise appear to be extremely thin plates of hematite-ilmenite lining the cleavage planes. The opaque phase associated with the serpentinized cracks in the pyroxene is hematite. No spinel phase was seen in reflected light.

The most likely origin for the primary oxide rodlets and plates in the plagioclase and pyroxene is thought to be exsolution from the silicate, although the crystallochemical relations are not clearly understood. The bulk composition of the lamellae intergrowths (·5 hem: ·5 ilm) implies that they must have exsolved from the silicate at a temperature above the crest of the hematite-ilmenite solvus (~900° C; Carmichael, 1961).

The monzonrite sample contains feldspar crystals consisting of a plagioclase core, zoning outward to mesoperthite (40 percent), hypersthene (40 percent), with subsidiary augite (5 percent), and about 15 percent opaques. These consist of about equal portions of magnetite and ilmenite, either in discrete crystals, or associated in very coarse, sandwich intergrowth.

Measurement of saturation magnetization (8 kilogauss) versus temperature (Js/T) curves, in air and in vacuum, on an anorthosite chip and a fragment of a plagioclase single crystal revealed identical Curie point inflections between 570° and 590°. Taken alone this result can be considered to indicate the presence of pure magnetite ($T_c \simeq 580°$). Titaniferous hematite, however, is the only ferrimagnetic phase observed microscopically in reflected light. From the hematite-ilmenite solvus (Carmichael, 1961), unmixed hematite can contain at least 10 percent ilmenite in stable solid solution at room temperature, and the Curie point of such a titaniferous hematite would be about 590° C, very similar to that of pure magnetite. The Curie point recorded on the anorthosite and plagioclase samples is therefore tentatively concluded to be due to the titaniferous hematite observed, and not to magnetite. It should be noted, however, that the hematite phase in any natural unmixed hematite-ilmenite intergrowth will always contain *at least* 10 percent of ilmenite still in solid solution. The Curie point will be 600° C or below, and can be misinterpreted to indicate the presence of magnetite.

The AF demagnetization curve of the monzonorite sample (5–3–1, Fig. 2) indicates a substantial unstable component, with coercivity less than 100 oersted, and a stable component of intensity approximately 1×10^{-4} emu/cc, similar to the anorthosites. It is noteworthy that the monzonorite contains in excess of five percent by volume of magnetite, whereas the anorthosites contain less than one percent of titaniferous hematite as the only observable ferrimagnetic phase. The relative unimportance of homogeneous coarse-grained magnetite as a carrier of stable remanent magnetism is well illustrated by this example.

DISCUSSION OF PALEOMAGNETIC RESULTS

The stable component of NRM in all samples is reversed (N-seeking pole upwards), inclined at approximately 160° to the present field. As a result, elimination of the soft (normal) viscous component resulted, in some samples, in an initial increase in RM intensity.

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The consistency of direction, both within and between all sites, is very good (Table 1). The remanent magnetism of the plagioclase and hypersthene megacryst samples was likewise indistinguishable from the associated anorthosite. One specimen of two from the feldspar crystal sample seemed to possess a more stable secondary component which required 300 oersted to erase, whereas the other gave stable directions after 100 oersted. The pyroxene megacryst specimens, on the other hand, possessed a strong but initially relatively unstable remanence; what remained after 300 and 500 oersted, however, was the strongest of Egersund samples measured. Although the megacrysts gave consistent magnetic data, we judge the ordinary anorthosite to be superior material for paleomagnetic study because of their higher microscopic coercivity.

There is no evident difference between the results for the Egersund-Ogna and the Haaland-Helleren massifs, and similarly, the monzonorite sample is indistinguishable from the associated anorthosite. This agreement is consistent with the hypothesis that all the rocks sampled are consanguineous and that the NRM was

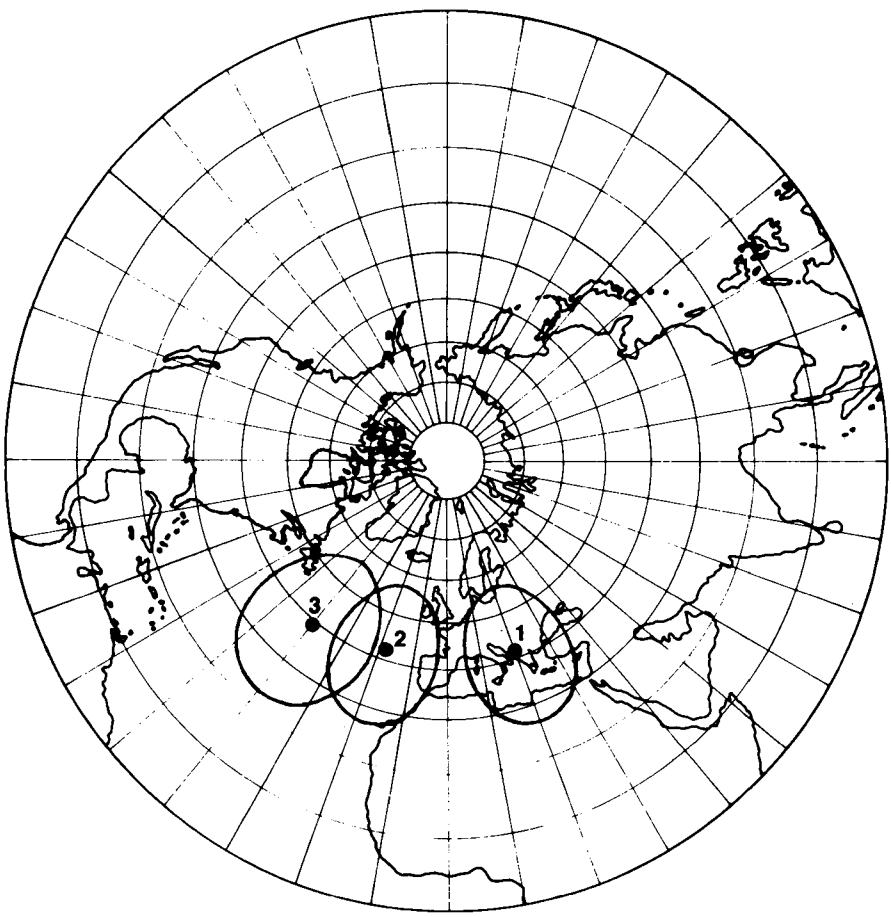


Figure 3. Northern hemisphere polar projection showing location of paleomagnetic poles. 1 = Egersund, this study; 2 = the Egersund pole rotated according to the continental reconstruction of Bullard and others (1965); 3 = Allard Lake (Hargraves and Burt, 1967).

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acquired simultaneously on cooling, being effectively "blocked" at a relatively low temperature ($\lessapprox 300°$ C).

It is assumed that this low temperature is comparable to that at which the diffusive loss of Ar^{40} ceased, that is, the best estimate of the time of acquisition of the remanent magnetism is that given by the K-Ar radiometric age. This is reported to be $\simeq$1,000 m.y. (Herz, 1969; Table 1). This pole is unlike that recorded from any European Paleozoic or younger rock, in keeping with the theory that it is a Precambrian paleomagnetism that has been preserved.

The Egersund anorthosite paleomagnetic pole is compared with that obtained from the Allard Lake, Quebec, anorthosite series (Hargraves and Burt, 1967) of similar 1,000 m.y. K-Ar age (Leech and others, 1963) as shown in Figure 3 and Table 2.

The poles are 45° apart when the Egersund pole is calculated according to the present-day coordinates of the sampling site. If the continents are juxtaposed according to the reconstruction of Laurasia given by Bullard and others (1965), however, the two are only 17° apart and are not significantly different (Watson and Irving, 1957; Table 2).

CONCLUSIONS

The results of the study are consistent with the hypothesis that the remanent magnetism of the anorthosite series at Allard Lake, Quebec, and Egersund, Norway, were acquired more or less contemporaneously approximately 1,000 m.y. ago, when the North American and European continents were contiguous, as in the Laurasian reconstruction. The original continuity of the "Grenville" belt from Quebec to Norway, as illustrated by Wynne-Edwards and Hasan (1970), is likewise favored by the data. The "proto-Atlantic Ocean," postulated by Wilson (1966) to have existed prior to the middle Paleozoic, must have, in turn, opened subsequent to the time of magnetization of the anorthosites.

TABLE 2. PALEOMAGNETIC POLES CALCULATED BY AVERAGING VIRTUAL POLES FOR EACH INDIVIDUAL SITE AND DATA FOR THE COMBINATION OF EGERSUND (ROTATED) PLUS ALLARD LAKE

Locality	Number of site poles averaged	Pole position	k	$\alpha 95$	R*
Egersund	5	42° N., 20° E.	28	15	4.86
Egersund (rotated)	5	42° N., 18° W.	28	15	4.86
Allard Lake (Hargraves & Burt, 1967)	4	39° N., 40° W.	27	18	3.89
Egersund & Allard Lake	9	41° N., 27.5° W.	24	10.8	8.66

*R = resultant vector; other symbols as in Table 1. A statistical test to determine whether the mean polar vectors for Allard Lake, and Egersund (rotated) are different (Watson and Irving, 1957) gave a result (see calculated parameters) indicating no significant difference at the 5-percent level:

$$\frac{N - 2(\Sigma Ri - R)}{N - \Sigma Ri} = 2.41$$

$$F_{2,2(N-2)}\ 5\% = 3.55.$$

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